

Evaluating the NICHE Program: Disparities in Aging Care Implementation Across US Hospitals

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Background



Addressing disparities in elderly care quality and accessibility, the Nurses Improving Care for Healthsystem Elders (NICHE) model¹ presents a transformative approach to geriatric nursing, equipping nurses for patient-centered, age-sensitive care, aiming to meet national quality objectives. Despite evidence of NICHE's effectiveness, **the scope and generalizability of its impact and the factors affecting implementation are not fully understood**. This study aims to (1) assess the socio-demographic distribution of US hospitals engaged with NICHE and (2) risk factors influencing initial care quality disparities and modifying the NICHE effect over time.

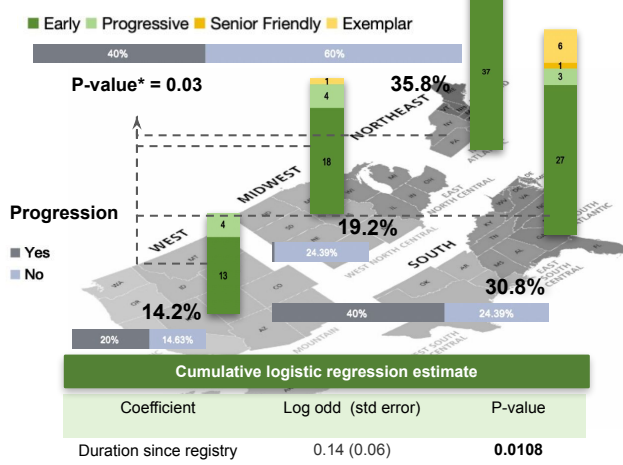
Methods

- This analysis included 120 registered hospitals from 2003 to 2023 with implementation level (early, progressive, senior-friendly, exemplar) evaluated by NICHE team and demo-social information.
- Initial implementation levels and sociodemographic data were described using univariate descriptive statistics to understand distribution of membership characteristics.
- Bivariate analyses and a cumulative ordinal regression model were utilized to assess the unadjusted and adjusted effect of factors on initial implementation level.
- The modifying effect was studied using per-post analysis on self-derived outcome, progression to higher level or not comparing their initial and latest evaluated level, on the subset of sample with more than 1 observations over time.

Results

Characteristics	Total (N=120)	Early (N=95)	Progressive (N=14)	Senior friendly (N=4)	Exemplar (N=5.8)	P-value*
Clinical assess (No)	117 (97.5%)	92 (96.8%)	14 (100%)	4 (100%)	7 (100%)	0.847
Rural Site (No)	114 (95.0%)	89 (93.7%)	14 (100%)	4 (100%)	7 (100%)	0.645
Bed size mean (sd)	(329, 316)	325 (338)	345 (198)	397 (317)	317 (213)	0.801
Duration since registry mean (sd)	(2.50, 4.22)	2.21 (4.31)	1.93 (2.53)	1.00 (0.816)	8.43 (1.99)	0.003

* P-value obtained by Kruskal-Wallis test to examine between-group differences across initial levels



* P-value obtained by Wald test

Results

- Only 5% hospitals are recorded from rural sites. Hospital participation was most prevalent in the Northeast and South regions.
- Hospitals with longer NICHE affiliations ended up with higher implementation levels. Particularly, each additional year of membership increased the odds of having higher implementation level by 15%.
- Hospitals in South had highest percentage to progress during their membership and end up at higher implementation level compared to other regions.

Implications

- **The study confirms NICHE's role in enhancing aging care.**
- Results from the **South hospitals suggest relatively better implementation**. Sources state that geriatric care is superior in other states, in spite of this the South had better implementation in the centralized health systems potential in standardizing care.
- However, **the study also underscores a critical urban-rural divide**, indicating the program's urban-centric reach. This gap signifies an essential need for extending NICHE's benefits to rural areas, where aging care services are needed. According to 2019 US Census Bureau report, 17.5% of rural populations were aged 65+ , compared to 13.8% of urban populations. (Source:census.gov)

Future work

- NICHE model, referred as a 'national' level program, should further reflect its national-targeted mission. Disparities in aging care through NICHE should be addressed at both urban and rural level. While maintaining its urban emphasis, NICHE may seek to disseminate and outreach its benefits to rural communities considering resource optimization through related stakeholder engagement.
- Future study should dive into discussing implementation framework embedding NICHE in rural settings. Innovative implementation strategies along with stakeholder engagement would facilitate extending this program for rural setup. For instance, all stakeholders should be acknowledged for their impactful work, being deliverers of quality care but also adding to existing medical knowledge through reporting data.
- Due to limited waves, the study only examined risk factors by conducting cross-sectional and pre-post analysis comparing 2 waves of observations. Future study should involve extended longitudinal research to further confirm the characteristics related to NICHE adoption, enabling thorough evaluations of patient outcomes.²

References

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